Q. ER single sequence is guided to the ER by which of the two components?

The ER signal sequence is guided to the ER by 2 components: 1) SRP which cycles between the ER membrane and the cytosol & binds to the signal sequence, & 2) SRP receptor in the ER membrane

Q. Write the name of antibody that appears on the surface of developing B cell?

IgM is the 1st class of Ab to appear on cell surface and secreted B cells can then switch Ab class

And IgD.

Q. What is the role of regulatory region explains its two elements.

Transcription Regulation:

• Some of genes are expressed in all cells all the time. Housekeeping genes are responsible for the routine metabolic functions (e.g. respiration). • Some are expressed as a cell enters a particular pathway of differentiation. • Some are expressed all the time in only those cells that have differentiated in a particular way. • Some are expressed only as conditions around and in the cell change. For example, the arrival of a hormone may turn on/off certain genes in that cell. Gene

Q. Describe the two signaling functions of N-terminal ER signal sequence of a soluble protein

The N-terminal ER signal sequence of a soluble protein has two signaling functions 1) It directs the protein to the ER membrane 2) It serves as a start-transfer signal (or start-transfer peptide) that opens the pore.

Q. What are the name of antibodies ?

5 classes of antibodies, IgA, IgD, IgE, IgG, & IgM, class of heavy chain—α, δ, ε, γ, & μ, respectively.

Q. Why Stem cells are different from specialized cells ?

Because Stem cells are unspecialized cells that renew themselves Under certain physiologic or experimental conditions, Stem cells can be induced to becodme cells with special functions.

Q. How UV helps in sterilization of equipment?

Sterilization, the killing of bacteria (or any types of cells), on medical instruments is primarily achieved by the radiation causing severe damage to the cell's components and to the cell's chromosomes, specifically the DNA. "Severe damage" to DNA is multiple breaks in the long DNA ladder-like structure. Radiation (for example, Ultra violet radiation) has enough energy to ionize atoms and molecules; that is, it can create charged particles and free radicals. In cells, the water can be ionized to form free radicals, somewhat like H-plus (H^+) and OH-minus (OH^-) radicals found in acids (for example H^+Cl^-) and bases (for example, Na^+OH^-). The free radicals made inside a cell's nucleus can cause damage to the cell's components and breaks in the DNA structure (the radiation may also directly cause ionizing breaks in

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the DNA's chemical chain). If enough damage is done to the cell's DNA, which acts as the cell's control mechanism, the cell cannot function properly or reproduce and the result is cell death (sterilization).

Q. What is stem cell and its benefits and therapeutic uses?

An undifferentiated cell of a multicellular organism which is capable of giving rise to indefinitely more cells of the same type, and from which certain other kinds of cell arise by differentiation.

Benefits

The hope going forward is that stem cells can also be used as a "renewable source of replacement cells and tissues" to treat common and serious diseases without the need for organ transplants or surgeries, including: macular degeneration, spinal cord injury, stroke, burns, heart disease, diabetes, osteoarthritis,

Therapeutic Uses

- Speed up the length of time it takes for injuries or wounds to heal
- Reduce pain, even **chronic joint pain**, with less need for medications
- Increase functionality, range of motion, flexibility and sleep quality
- Reduce muscle compensations and risk for future injuries
- Decrease nerve damage
- Increase collagen
- Help generate new heart and blood vessel tissue
- Help heal skin wounds, prevent formation of scar tissue and **reduce hair loss**
- Return patients to their normal activities as quickly as possible

Q. Describe three steps of RNA processing?

Pre-mRNA (freshly transcribed) is processed to make it functional.

1) Introns are removed during RNA maturation
2) The transcribed pre-mRNA is altered by the addition of a G cap modified GTP at the 5′ end to protect RNA & facilitate translation.
3) After the last codon a poly A tail (100-300) is added at the 3′ end for export and protection

Q. What is mechanism of the olfaction receptors..?

Humans can distinguish more than 10,000 distinct smells, which are detected by specialized olfactory receptor neurons in the lining of the nose. These cells recognize odors by means of specific G-protein-linked olfactory receptors, which are displayed on the surface of the modified cilia that extend from each cell. The receptors act through cyclic AMP. When stimulated by odorant binding, they activate an olfactory-specific G protein (known as Golf), which in turn activates adenylyl cyclase. The resulting increase in cyclic AMP opens cyclic-AMP-gated cation channels, thereby allowing an influx of Na+, which depolarizes the olfactory receptor neuron and initiates a nerve impulse that travels along its axon to the

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brain. It can detect thousands of different odor molecules through a large family of olfactory receptors (ORs), of a diverse protein sequence, which are located in olfactory sensory neurons (OSNs) in the olfactory epithelium in the nose of humans, and in the vomeronasal organ in animals.

Mechanism

Odorant molecules pass through the nasal cavity.

1: binding of odorant to its receptor activates G proten

2: the activation of G protein activates the synthesis of cAMP by adenylyl cyclase.

3: cAMP activates the opening site of ion channel

4: change in the ion concentration inside the cell send a signal to a specific area of the brain perceives the signal as a scent.

Q. Elaborate the mechanism of growth promoting mechanism of oncogenes?

Growth Promoting Oncogenes:

a) Growth Factors b) Growth Factor Receptors c) Signal-Transducing Proteins (secondary messengers) d) Nuclear Transcription Factors e) Cyclins and Cyclin Dependent Kinases (CDKs)

Autonomous Cell Growth In Cancer Cells Is Promoted By Following Elements

A) Growth Factor

Some cancer cells acquire the ability through mutations to produce growth factors to which they are responsive (autocrine signaling). Many glioblastomas secrete PDGF and sarcomas make TGF- α. Similar loops are fairly common many types of cancers.

B) Growth Factor Receptors

Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand. EGF receptor ERBB1 is over expressed in 80% of SCC of lung. ERBB2/HER2 is amplified in 25%-30% of breast cancers and adenocarcinomas of lung , ovary and salivary glands. These tumours are sensitive to small amounts of growth factors.

C) Signal-Transducing Proteins (Secondary Messengers)

Mutaion of RAS is the most common oncogene abnormality, approximately 30% of all human tumors contain mutated version of RAS gene. GAP/neurofibramin 1 (a RAS deactivating protein) is mutated in neurofibramatosis type 1. BCR-ABL hybrid gene activates RAS pathways (like p53 Normal ABL gene also promotes apoptosis via NLS domain).

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D) Nuclear Transcription Factors

MYC and the polymerase assemble on a specific DNA sequence and drive transcription of genes. MYC protooncogene is expressed in virtually all the cells, & the MYC protein is rapidly induced when cells receive a signal to divide. MYC binds DNA and causes transcriptional activation of many growth related proteins. Dysregulation of of MYC gene occurs in Burkitt Lymphoma (B-cell tumor); MYC is also amplified in many other cancers

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Q. How G protein involved in formation of GDP? 2 marks

When Rhodopsin absorbs the light (Photon) causing the G protein .transduction to exchange GTP for GDP.

Q. How sheep is cloned by the nuclear material. Explain it by all steps. 5 marks

Nuclear transfer is a form of cloning. The steps involve removing the DNA from an oocyte (unfertilised egg), and injecting the nucleus which contains the DNA to be cloned. In rare instances, the newly constructed cell will divide normally, replicating the new DNA while remaining in a pluripotent state. If the cloned cells are placed in the uterus of a female mammal, a cloned organism develops to term in rare instances. This is how Dolly the Sheep and many other species were cloned. Cows are commonly cloned to select those that have the best milk production.

Despite this, the low efficiency of the technique has prompted some researchers, notably Ian Wilmut, creator of Dolly the cloned sheep, to abandon it

Q. Write synapse and its steps involved in the mechanism. 5 marks

a junction between two nerve cells, consisting of a minute gap across which impulses pass by diffusion of a neurotransmitter.

Steps:

There are two general classes of neurotransmitters: large neuropeptides or smaller amines/amino acids which are synthesized differently.

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Step 1 – Neurotransmitter Synthesis. ...

Step 2 & 3- Neurotransmitter Packaging and Release. ...

Step 4 – Neurotransmitter Binding. ...

Step 5 – Stopping the Chemical Signal.

Mechanism

Neurotransmitters, such as Acetylcholine, are stored in vesicles in the presynaptic terminal. Acetylcholine is secreted across synaptic clefts, where it binds to the post synaptic membrane and causes a change in the permeability of the postsynaptic cell, which results in an excitatory post-synaptic potential

Q. Describe TGF-ß Receptor Family.

Consists of a large number of structurally related, secreted, dimeric proteins. During development, they regulate pattern formation and influence various cell behaviors, ie proliferation, differentiation, extracellular matrix production, and cell death. In adults, tissue repair and in immune regulation. Act through enzyme-linked receptors that are single-pass transmembrane proteins. Ligand first binds and activates a type-II receptor homodimer, which recruits, phosphorylates, and activates a type-I receptor homodimer, forming an active tetrameric receptor complex Activated TGF-β receptors phosphorylate Smad2 activated BMP receptors phosphorylate Smad1, Smad5, or Smad8.One of these Smads then dissociates from the receptor and binds to Smad4. Both type-I and typeII receptors, required for signaling. The type-I receptor directly binds and phosphorylates a latent gene regulatory protein of Smad family

Q. Define Electronegativity ?.

The tendency of an atom to attract electrons towards itself.

Q. What is Chargaff rule?

In DNA, the amount of Purines (A+G) is always equal to the amount of pyrimidine (T+C)

Q. Write mitochondrial features?

Mitochondria are double-membrane-enclosed organelle. It contains its own DNA, ribosomes, and other components required for protein synthesis, most of its proteins are encoded in the cell nucleus and imported from the cytosol There are two subcompartments in mitochondria: the internal matrix space and the intermembrane space Proteins imported into the matrix of mitochondria are usually taken up from the cytosol within seconds or minutes of their release from ribosomes. Mitochondrial proteins are first fully synthesized as precursor proteins in the cytosol & then translocated into mitochondria by a posttranslational mechanism.

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Q. Write Briefly note on cell cycle?

The cell cycle consists of four distinct phases: G_1 phase, S phase (synthesis), G_2 phase (collectively known as interphase) and M phase (mitosis). M phase is itself composed of two tightly coupled processes: karyokinesis, in which the cell's chromosomes are divided, and cytokinesis, in which the cell's cytoplasm divides forming two daughter cells. Activation of each phase is dependent on the proper progression and completion of the previous one. Cells that have temporarily or reversibly stopped dividing are said to have entered a state of quiescence called G_0 phase.

Interphase

Before a cell can enter cell division, it needs to take in nutrients. All of the preparations are done during interphase. Interphase is a series of changes that takes place in a newly formed cell and its nucleus, before it becomes capable of division again. It is also called preparatory phase or inter-mitosis. Previously it was called resting stage because there is no apparent activity related to cell division. Typically interphase lasts for at least 90% of the total time required for the cell cycle.

Interphase proceeds in three stages, G_1, S, and G_2, followed by the cycle of mitosis and cytokinesis. The cell's nuclear DNA contents are duplicated during S phase.

G_1 phase (growth)

The first phase within interphase, from the end of the previous M phase until the beginning of DNA synthesis, is called G_1 (G indicating *gap*). It is also called the growth phase. During this phase, the biosynthetic activities of the cell, which are considerably slowed down during M phase, resume at a high rate. The duration of G_1 is highly variable, even among different cells of the same species.[3] In this phase, the cell increases its supply of proteins, increases the number of organelles (such as mitochondria, ribosomes), and grows in size.

S phase (DNA replication)

The ensuing S phase starts when DNA synthesis commences; when it is complete, all of the chromosomes have been replicated, i.e., each chromosome has two (sister) chromatids. Thus, during this phase, the amount of DNA in the cell has effectively doubled, though the ploidy of the cell remains the same.

G_2 phase (growth)

G2 phase occurs after DNA replication and is a period of protein synthesis and rapid cell growth to prepare the cell for mitosis. During this phase microtubules begin to reorganize to form a spindle.

Mitotic phase (chromosome separation)

The relatively brief *M phase* consists of nuclear division (karyokinesis). It is a relatively short period of the cell cycle. M phase is complex and highly regulated. The sequence of events is divided into phases, corresponding to the completion of one set of activities and the start of the next. These phases are sequentially known as:

- prophase, metaphase, anaphase, telophase

Q. Define Genetic engineering?

Genetic engineering is the process of manually adding new DNA to an organism. The goal is to add one or more new traits that are not already found in that organism.

Q. What is Incomplete Dominance?

Incomplete dominance is a form of intermediate inheritance in which one allele for a specific trait is not completely expressed over its paired allele. This results in a third phenotype in which the expressed physical trait is a combination of the phenotypes of both alleles. Example 4 O'clock

Q. Define Codominance?

A situation, when both alleles appear in the phenotype neither allele is dominant. Both alleles are expressed in heterozygous individuals. Example Blood Group

Q. Define Mutation?

A **mutation** is a change that occurs in our DNA sequence, either due to mistakes when the DNA is copied or as the result of environmental factors such as UV light and cigarette smoke

Q. Define Epistasis?

Effect of one gene dependent on the presence of one or more 'modifier genes'

Q. What is Aquaporin?

Water molecules move through membranes by Osmosis (diffusion).

Q. What are PCR applications?

1) DNA Fingerprinting 2) Genotyping 3) Genomic Cloning 4) Gene expression studies 5) Drug discovery 6) classification of organisms etc.

Q. Write name of 5 antibiotics?

1) Tetra cycline 2) penicillin 3) amoxicillin 4) erythromycin 5) levofloxacin

Q. Define epinephrine and how it affected by beta blocker?

A hormone that is secreted by the adrenal gland in response to physical or mental stress, as from fear, and is regulated by the autonomic nervous system. The release of **epinephrine** causes an increase in heart rate, blood pressure, and respiratory rate. β-blockers Bind to the receptor but do not activate hormone induced effects.

Q. What are DNA replication Steps?

The three steps in the process of DNA replication are 1) initiation 2) elongation 3) termination.

Q. elaborates points between merged cAMP activity? 2 marks

cAMP-dependent protein kinases. Phosphorylate specific serine & threonine residues. Phosphorylation of many enzymes increases their catalytic activity, whereas phosphorylation of others decreases their activity.

Q. Define codon? 2 marks

A sequence of three nucleotides which together form a unit of genetic code in a DNA or RNA molecule called codon.

Q. Difference b/w klinefelter syndrome and turner syndrome.? 2 marks

Klinefelter syndrome is a genetic **disorder** that affects males. **Klinefelter syndrome** occurs when a boy is born with one or more extra X chromosomes (47 XXY)

Turner syndrome is a chromosomal condition in females that lack one X chromosomes they contain 45 chromosomes.

Q. How turbutalene and b blocker affect the function of epinephrine adrenaline.? 5 marks

β1- and β2-adrenergic receptors (Heart & Lungs Respectively) are coupled to G proteins (Gs), which activate adenylyl cyclase. α1 and α2 receptors are coupled to two other G proteins, Gq and Gi, respectively. adenylyl cyclase Inhibition & phospholipase C activation. Epinephrine bound to α2 receptors causes the arteries to constrict, cutting off circulation to the blood vessels in the intestinal tract, skin, and kidneys. (peripheral organs) Agonists (terbutaline) Mimic the function of a hormone by binding to its receptor and causing the normal response. Antagonists (β-blockers) Bind to the receptor but do not activate hormone induced effects. An antagonist acts as an inhibitor of the natural hormone (or agonist) by competing for binding sites on the receptor, thereby blocking the physiological activity of the hormone.

Q. Difference b/w confluent monolyer and post confluency stages?. 5 marks

A confluent monolayer is an adherent cell culture (dish, plate or flask) in which the cells have formed a single layer over the entire surface area available for growth. • Once the cells have started to form clusters above the first layer or have started to lift up from the surface, the culture is past confluency stage. • Cells that are repeatedly passaged at densities past confluency display decreased doubling times, decreased viabilities, & decreased ability to attach.

Q. Describe the larger subunit of ribosomes? 5 marks

T (transfer) site: first site where charged tRNA lands with the help of transfer factor A (AA) site: Anticodon binds to mRNA codon, P (polypeptide) site: tRNA adds AA to polypeptide, E (exit) site: after AA transfer tRNA exits here.

1) A charged tRNA with anticodon to next codon binds A site.

2) Large subunit breaks bond between tRNA & its AA at P site.

3) Assisted by elongation factors large subunit Catalyzes formation of peptide bond b/w the AA & the one attached to tRNA in A site (peptidyl transferase activity).

Q. Define carbohydrates and discuss it's classes? 3

Carbohydrates are polyhydroxy aldehyde or polyhydroxy ketones. It is further classified into 4 major groups.

The simplest **monosaccharides** are the two three-carbon trioses

Examples : – Glyceraldehyde,an aldotriose – Dihydroxyacetone, a ketotriose

Disaccharides are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis – Maltose - 2 molecules of glucose on hydrolysis

Examples. Maltose , sucrose , galactose

Oligosaccharides are short chains of monosaccharide units or residues (310) joined by characteristic linkages called glycosidic linkage

Polysaccharides (Glycans) are those which yield more than 10 molecules of monosaccharides on hydrolysis • Some have hundreds or thousands of units • Some polysaccharides, such as cellulose, are linear chains; others, such as glycogen, are branched

Q. What do you know about endocytosis? Differentiation between phagocytosis and pinocytosis?

The material to be ingested is progressively enclosed by a small portion of the plasma membrane, which first invaginates and then pinches off to form an endocytic vesicle containing the ingested substance or particle

Phagocytosis involves the ingestion of large particles microorganisms or dead cells via large vesicles called phagosomes (>250 nm in diameter)

Pinocytosis involves ingestion of fluid & solutes via small pinocytic vesicles (about 100 nm in diameter)

Q. What are missense mutations? (2 marks)

Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein

Q. What are the types of homotypic and hetrotypic and how they formed?

After transport vesicles have budded from an ER exit site they shed their coat, and then they begin to fuse with one another this fusion of membranes from the same compartment is called **homotypic fusion**, to distinguish it from **heterotypic fusion**, in which a membrane from one compartment fuses with the membrane of a different compartment.

Q. What is TRANSMISSION OF SINGNAL FROM SYNAPSIS TO NEUROMUSCLE?

A **neuromuscular** junction (or myoneural junction) is a chemical **synapse** formed by the contact between a motor neuron and a muscle fiber. It is at the **neuromuscular** junction that a motor neuron is able to **transmit** a **signal** to the muscle fiber, causing muscle contraction.

Q. What is NEGATIVE and Positive CONTROL OF TRANSCRIPTION.?

The regulatory molecule (the repressor) prevents transcription. Negative Control of Transcription:

Positive Control of Transcription: the regulatory molecule (the CRP–cAMP complex) enhances transcription.

Q. Write ADULT STEM CELL AND ROLE?

Adult Stem cells Are harvest from there niches from a (postnatal) individual,No ethical dilemma for their use in research, Have a limited ability to renew themselves and differentiate in to different cell types. Do not cause an immune rejection.

Q. What is G proteins vision 2

Activates cyclic GMP phosphodiesterase in vertebrate rod photoreceptors.

Q. Write Role of MHC 2

The function of MHC molecules is to bind peptide fragments derived from pathogens and display them on the cell surface for recognition by the appropriate T cells.

Q. Difference of Transcription in prokaryotes & Eukaryotes3 marks

Prokaryotic Transcription	**Eukaryotic Transcription**
Occurs in the cytoplasm	Occurs in the Nucleus
There is no definite phase for its occurrence.	take place in the G1 and G2 phases of cell cycle
initiation of transcription does not need any proteins or initiation factors	initiation of transcription requires proteins called transcription factors like TFIIA, TFIIB , TATA BOX etc
A single RNA Polymerase synthesises all the three types of RNA(mRNA, tRNA, rRNA)	The RNA Polymerase I, II and III synthesizes rRNA, mRNA and tRNA respectively
The 23S, 16S and 5S rRNAs are formed from a single primary transcript	The 28S, 18S, 58S and 5S rRNAs are formed from two primary transcripts

Q. Write a note on cloning 5 marks

The term in itself means the *reproduction of a biological entity which is genetically identical to the one from which it originated.* The Greek term *'klon'* makes us think of a sprig from a plant which when inserted into mature soil can reproduce the plant from which it was taken.

Steps of Cloning

1) the reproduction of a cell line starting from a single cell: the cells produced are histologically identical to the ones from which they originated. Cloning is also spoken of for the reproduction of DNA fragments starting from a single DNA fragment;

2) the reproduction of embryos by disaggregating or subdividing an embryo (*embryo splitting*) in its early stages of development when the cells are totipotent or pluripotent, i.e., capable of developing into a complete organism. The embryos obtained are then transferred *in utero*;

3) the reproduction of genetically identical individuals through the *nucleo-transfer* of a somatic cell from an embryo, fetus or an adult individual, to a denucleated oocyte. The embryos obtained are later transferred *in utero*. The procedure can be repeated with several nuclei taken from the somatic cells of the same individual and inserted into denucleated ova;

4) the transfer of the nucleus of an oocyte to the cytoplasm of a second denucleated oocyte, as a form of "prevention" of mitochondrial diseases. This, however, is not cloning in a strict sense. The modified oocyte is later fertilized in vitro and transferred in utero.

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Q. Write Types of mutations 5 marks

Somatic mutations: Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

Germ line mutations: Occurs in the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

Point mutations: mutations of single base pairs.

Chromosomal mutations: change the position or orientation of a DNA segment.

Silent Mutations: Mutations result in no change in amino acids due to redundancy genetic code

Nonsense mutations: Base substitution results in stop codon, (UAG) to form in the mRNA product

Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein

Frame-Shift Mutations: No substitutions. Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed

Q. What is Nuclear pore complex 10 marks

Eukaryotic nuclear envelope is perforated by large, elaborate structures called nuclear pore complexes which selectively facilitate nuclear transport. Each pore complex contains one or more open aqueous channels through which small water-soluble molecules can passively diffuse. Each complex has a molecular mass of about 125 million & is composed of aprox 50 different proteins, called nucleoporins, arranged octagonal symmetry. Small molecules (5000 daltons or less) diffuse in so fast that the nuclear envelope can be considered to be freely permeable to them, a protein of 17,000 daltons takes 2 minutes to proteins larger than 60,000 daltons are unable to enter the nucleus. Nuclear pore contains a pathway for free diffusion 9 nm in diameter & 15 nm long, this channel occupies a small fraction of total volume of the pore complex. The more active the nucleus is in transcription, the greater the number of pore complexes .The nuclear envelope of a typical mammalian cell contains 3000–4000 pore complexes. Cells synthesizing DNA, need to import about 106 histone molecules from the cytosol every 3 minutes to package DNA into chromatin, ie each pore complex needs to transport about 100 histone molecules/minute. Rapidly growing cell need to transport about 6 large & small ribosomal subunits/minute from nucleus, where they are produced, to the cytosol

Q. What is Complement system 10 marks

Complement System. Most of the time, the binding of antibodies to antigen performs no useful function until and unless it can activate an effector mechanism. Complement system serves several effector roles. Complement system provides the actual protection from the response while the interaction of antibodies and antigen provides the specificity of the response. Put another way, antibodies "finger" the target, complement destroys it the complement system consists of some 30 proteins circulating in blood plasma. These proteins are synthesized mainly in the liver, and they account for about 5% of the globulin fraction of blood serum. Most of these are inactive until .they is cleaved by a protease which, in turn, converts them into a protease. Thus many components of the system serve as the substrate of a prior component and then as an enzyme to activate a subsequent component. The classical pathway is

triggered by activation of the C1-complex (which consists of six molecule C1q and two molecules C1r and C1s), either by C1q's binding to antibodies from classes M and G, complexed with antigens, or by its binding C1q to the surface of the pathogen. C4b, which binds covalently to sugar residues on cell-surface glycoproteins,& a smaller, inactive, fragment of C4a which diffuses away C2 is cleaved into C2b, which binds non-covalently to a site on C4b, leaving a smaller, inactive, fragment of C2a which diffuses away

This binding leads to conformational changes in C1q molecule, which leads to the activation of two C1r (serine protease) molecules. Then they cleave C1s. (another serine protease) The C1-complex now binds to and splits C2 and C4, producing C2a and C4b

The complex of C4b•2b is called "C3 convertase" because it catalyzes the cleavage of C3 C3b binds covalently to glycoproteins scattered across the cell surface. Macrophages and neutrophils have receptors for C3b and can bind the C3b-coated cell or particle.

Q. Differentiate between pyrimidine and purines?

Purine is a double ring nitrogenous base while pyrimidine is a single ring nitrogenous base.

Purine contains two nitrogenous bases i.e Adenine and guanine while pyrimidine contains three nitrogenous bases i.e cytosine, thymine and uracil

Q. Enlist requirement for PCR?

1) Target DNA 2) Two oligonucleotide primers 3) DNA polymerase, 4) dNTPs 5) PCR machine

Q. DNA replication?

In molecular biology, DNA replication is the biological process of producing two identical replicas of DNA from one original DNA molecule. This process occurs in all living organisms and is the basis for biological inheritance. DNA is made up of a double helix of two complementary strands.

Q. Explain DNA replication Models?

Conservative replication: would preserve the original molecule and generate and entirely new molecule.

Semi conservative Replication: would produce molecules with both old and new DNA but each molecule would contain one complete old strand and one new one.

Dispersive Replication: would produce two molecules with old and new DNA interspersed along each strand

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Q. Function restriction endonucleases

Special bacterial enzyme can cut double stranded DNA at specific sites in a test tube. Restriction endonucleases (restriction enzymes) act as a kind of immune system, protecting the cell from the invasion of foreign DNA (virus).

Q. Difference between DNA and RNA? (5)

DNA	RNA
It is double stranded molecule	It is a single stranded molecule
Cytosine pairs with Thymine	Cytosine pairs with Uracil
It contains DeoxyRibose sugar	It contains Ribose sugar
It replicates to form new DNA	It cannot replicates itself
It is long lived molecule	It has short life span except a few RNAs

Q. Describe process of mitosis?(5)

Mitosis is a type of cell division in which one parent cell divides to form two daughter cells that contain same no of chromosomes this type of cell division occurs only in somatic cells. It has phases

1) Prophase: The beginning of mitosis. so the individual chromatids become visible. Microtubules form between each pole and the chromosomes to make up a spindle.
2) Prometaphase: is marked by the disappearance of the nuclear envelope. Chromosomes begin to move toward the poles.
3) Metaphase: All the centromeres arrive at the equatorial plate. Metaphase is the best time to see the sizes and shapes of chromosomes because they are maximally condensed.
4) Anaphase: Separation of the chromatids marks the beginning of anaphase, during which the two sister chromatids move to opposite ends of the spindle. Each chromatid con-tains one double-stranded DNA molecule and is now referred to as a daughter chromosome.
5) Telophase: Chromosomes stop moving at the end of anaphase, the cell enters telophase. The chromosomes begin to uncoil, the nuclear envelopes and nucleoli, reform their respective structures

Q. Gel electrophoresis

A gel is made up of agarose polymer suspended in a buffer. it sits in a chamber between two electrodes, depression in the gel (wells) are filled with DNA solution, restriction enzyme 1 cuts the DNA once resulting in fragments A and B. Restriction Enzyme 2 cuts the DNA once, at a different restriction sequences. if other restriction enzymes are used , two cuts are made in the DNA. Each sample is loaded into one well in the gel. as fragments of DNA move toward the positive electrode , shorter fragments move faster than longer fragments. The fragments can be identified by gel electrophoresis on the basis of their sizes.

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Q. Quaternary structure of protein

Protein **quaternary structure** is the number and arrangement of multiple folded protein subunits in a multi-subunit complex. It includes organizations from simple dimers to large homooligomers and complexes with defined or variable numbers of subunits.

Q. Define Tautomer

Tautomers are constitutional isomers of organic compounds that readily interconvert. The concept of tautomerizations is called tautomerism. The chemical reaction interconverting the two is called tautomerization.

Q 132 Write functions of fibriogen 2 marks

Fibrinogen, or factor I, is a blood plasma protein that's made in the liver.**Fibrinogen** is one of 13 coagulation factors responsible for normal blood clotting. When you start to bleed, your body initiates a process called the coagulation

Q. What is Alpaktonuria?

Alkaptonuria (black urine disease, black bone disease, or alcaptonuria) is a rare inherited genetic disorder in which the body cannot process the amino acids phenylalanine and tyrosine, which occur in protein.

Q. Write three stop codons?

1) UAA 2) UAG 3) UGA

Q. Write three RNA Viruses?

HCV , HBV , HIV , Adeno Viruses, INfluenza Virus , cytomegalo Virus etc

Q. How Different types of bacteria will produce

Binary fission ("division in half") is a kind of asexual reproduction. It is the most common form of reproduction in prokaryotes specially bacteria reproduce by binary fission.

Q. F. Griffith was used in his experiment? (2 marks

Griffith used two strains of Pneumococcus bacteria, type III-S (smooth) and type II-R (rough)

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Q. Differentiated between B-Cells and T-Cells

B- Lymphocytes (B-Cells)	T-Lymphocytes (T-cells):
1. They arise from bone marrow Bursa of Fabricus	1. They arise from bone marrow Thymus.
2. B-cells form humoral or antibody-mediated immune system (AMIS).	2. T-cells form cell-mediated immune system (CSM).
3. They defend against viruses and bacteria that enter the blood and lymph	They defend against pathogens including protists and fungi that enter the cells.
4. They are formed by the division plasma cells	4. They are formed by the division of lymphoblasts of three types: killer, helper and suppressor cells.
5. Plasma cells do not move to the site of infection.	5. Lymphoblasts move to the site of infection.
6. Plasma cells do not react against transplants and cancer cells.	6. Killer cells react against transplants and cancer cells.

Q. Define MHC and Differentiates between MHC-I and MHC-II. (10 Marks)

The function of MHC molecules is to bind peptide fragments derived from pathogens and display them on the cell surface for recognition by the appropriate T cells.

MHC I	MHC II
Present on the surface of all nucleated cells	Present on the surface of antigen presenting cells such as macrophages, B cells, dendritic cells etc.
Class I is a single chain (alpha chain) bound to an invariant partner	class II is a heterodimer containing two roughly identical chains (alpha and beta chains)
Class I present 8-10 amino acid peptides	class II presents 14-18 amino acid Peptides
Mainly interact with cytotoxic T cells	Mainly interact with helper T cells

Encoded by HLA-A, HLA-B and HLA-C genes	Encoded by HLA-D gene
Bind endogenous antigens synthesized in a cell	Binds exogenous antigens
Presence of foreign or over-abundant antigens targets cell for destruction	Presence of foreign antigens induces antibody production, and attracts immune cells to area of infection

Q. what is RFLPs and its function

What is RFLP: RFLP is a difference in homologous DNA sequences that can be detected as fragments of different lengths after digestion of the DNA samples.Most RFLP markers are heterozygous and highly locus-specific. RFLP probe is a labeled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis

Q. Difference between homozygous and heterozygous

A homozygous trait is when two of the same kind of alleles combines to form a trait. Eg. TT or tt

Heterozygous is a trait when two different kinds of alleles combine to form a trait. Eg Tt

Q. Difference between Stem cells and Specialized Cells

Stem cells differ from other kinds of cells in the body. All stem cells—regardless of their source—have three general properties: they are capable of dividing and renewing themselves for long periods; they are unspecialized; and they can give rise to specialized cell types

Groups of **specialized cells** cooperate to form a tissue, such as a muscle. Different tissues are in turn grouped together to form larger functional units, called organs. Each type of cell, tissue, and organ has a distinct structure and set of functions that serve the organism as a whole

Q. CaM kiase II role in movement of muscular tissue

Myosin light-chain phosphatase, more commonly called myosin phosphatase (EC 3.1.3.53), is an enzyme (specifically a serine/threonine-specific protein phosphatase) that dephosphorylates the regulatory light chain of myosin II. This dephosphorylation reaction occurs in smooth muscle tissue and initiates the relaxation process of the muscle cells. Thus, myosin phosphatase undoes the muscle contraction process initiated by myosin light-chain kinase. The enzyme is composed of three subunits: the catalytic region (protein phosphatase 1, or PP1), the myosin binding subunit (MYPT1), and a third subunit (M20) of unknown function. The catalytic region uses two manganese ions as catalysts to

dephosphorylate the light-chains on myosin, which causes a conformational change in the myosin and relaxes the muscle. The enzyme is highly conserved and is found in all organisms' smooth muscle tissue. While it is known that myosin phosphatase is regulated by rho-associated protein kinases, there is current debate about whether other molecules, such as arachidonic acid and cAMP, also regulate the enzyme.

Q. What is the role of rod cell in vision?

Rod cells are photoreceptor cells in the retina of the eye that can function in less intense light than the other type of visual photoreceptor, cone cells. Rods are usually found concentrated at the outer edges of the retina and are used in peripheral vision. Activation of photopigments by light sends a signal by hyperpolarizing the rod cell, leading to the rod cell not sending its neurotransmitter, which leads to the bipolar cell then releasing its transmitter at the bipolar-ganglion synapse and exciting the synapse.

Q. What is difference between Isotonic, Hypertonic and Hypotonic?

Isotonic	Hypertonic	Hypotonic
Isotonic solutions are two solutions that have the same concentration of a solute.	Hypertonic solution is one of two solutions that have a higher concentration of a solute.	Hypotonic solution is one of two solutions that have a lower concentration of a solute.

Q. Define Laminar flow cabinet principle and function?

Function: Provides sterile working area where cells can be handled without worrying about contamination.

Principal: In a **laminar flow hood** the air is passed through a HEPA (High Efficiency Particulates Air) filter which removes all airborne contamination to maintain **sterile** conditions. ... Now the **sterile** air **flows** into the working (flasking) area where you can do all your flasking work without risk of contamination.

Q. Write steps involving E-coli signaling

After ligand binding change in tertiary structure of extra cellular part of EnvZ leads to structural change in its cytoplasmic domain making it a kinase (auto..). EnvZ-P can now phosphor-ilate OmpR (responder) outside signal in and amplified. Receptor conformational change after ligand binding which activates kinase activity. Phosphorilation alters responder function. Signal amplified. Transcription factor activated. Protein synthesis results in altered cell activity.

Ligand binding causes a structural change permitting G protein to bind receptor. Binding of G protein to activated receptor causes it to exchange GDP for GTP (ligand releases).

Q. How antibody help the surface to produce B cells

Each B cell lymphocyte produces one unique antibody against one unique epitope. Once antibodies with sufficient specificity to the epitope can be encoded, the B cell begins to release antibodies into the bloodstream. These antibodies then bind specifically with the foreign molecule and allow the immune system to eliminate the molecule from the system.

Q. Define Gene linkage? (2)

Linkage is defined genetically as the failure of two genes to assort independently. Linkage occurs when two genes are close to each other on the same chromosome

Q. Define Palindromic Sequences?

Palindromic Sequences: A palindromic sequence is the same when read in 5′ to 3′ direction on either complementary strand of DNA. Example DENNIS SINNED GTAATG VS GTATAC

Q. VNTRs stand for?

Variable Number Tandem Repeats

Q. What are the steps of initiation of translation process?

a) **Shine Dalgarno sequence:**

In prokaryotes the rRNA of the small ribosomal subunit first binds to a complementary ribosome binding site upstream start codon AGGAGG this is called shine Dalgarno sequence on the mRNA.

b) Anticodon of methioninie charged to tRNA and bindes to start the codon (Methionine can be removed after translation).

c) In the small subunit rRNA stabilizes H-bonding between t and m RNA.

d) Large subunit facilitated by initiation factors binds the complex with methhionine charged tRNA at the P site.

Q. What are elongation steps in translation process?

a) A charged tRNA with anticodon to next codon binds A site.
b) Large subunit breaks bond between tRNA & its AA at P site.
c) Assisted by elongation factors large subunit Catalyzes formation of peptide bond between the AA and the one attached to tRNA in A site (peptidyl transferase activity).

Q. What are termination steps in translation process?

a) When stop codon enters A site elongation is terminates.

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b) Stop codons bind release factor, which hydrolysis bond between polypeptide chain and tRNA at P site.
c) Protein folding information is a function of AA sequence.

Q. Define Somatic mutations?

It Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

Q. Define Germ line mutations?

It Occurs the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

Q. What are Point mutations?

These are Mutations of single base pairs.

Q. Write function of chromosomal mutations?

It change the position or orientation of a DNA segment.

Q. Define silent mutation?

The Mutations which results in no change in amino acids due to redundancy genetic code.

Q. Define Missense Mutations with example?

These are Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein. For example Sickle-cell disease results from a defect in human β-globin gene, a protein in human red blood cells that carries oxygen.

Q. Define Nonsense mutations?

Base substitution results in stop codon, (UAG) to form in the mRNA product

Q. Define Frame-Shift Mutations?

In which No substitutions accurs a Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed.

Q. What is epigenetics?

Epigenetics is the study of heritable changes in gene expression that do not involve changes in the DNA sequence.

Q. Define operon?

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Cluster of genes with 1 promoter is called operon Operon consists of a promoter, an operator, and two or more structural genes.

Q. Differentiate between Positive Control of Transcription and negative Control of Transcription?

Positive vontrol of Transcription is the trascription in which regulatory molecule (the CRP–cAMP complex) enhances transcription. While negative Control of Transcription In which The regulatory molecule (the repressor) prevents transcription.

Q. Write Three Systems of regulations of operon?

- An inducible operon regulated by a repressor protein
- A repressible operon regulated by a repressor protein
- An operon regulated by an activator protein

Q. Write two essential sequence of the promoter?

Promoter has two essential sequences. 1) Recognition sequence (RNA polymerase). 2) TATA box where DNA begins to denature.

Q. How pre-mRNA (freshly transcribed) is processed to make it functional?

1-Introns are removed during RNA maturation. 2) The transcribed pre-mRNA is altered by the addition of a G cap (modified GTP) at the 5′ end to protect RNA and facilitate translation. 3) After the last codon a poly A tail (100-300) is added at the 3′ end for export and protection.

Q. What is magnetism?

Magnetism can be a Signal which in some bacteria and birds orient themselves to Earth's magnetic poles.

Q. Define Signal Transduction?

The entire signaling process, from signal detection to final response is called signal transduction pathway/ signaling Cascades

Q. What are types of protein in signaling pathway?

1) Relay proteins: pass the message to the next signaling component in the chain.

2) Messenger proteins: carry the signal from one part of the cell to another (cytosol to nucleus).

3) Adaptor proteins: link one signaling protein to another, without themselves conveying a signal.

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4) Amplifier protein: either enzymes or ion channels, increase the signal they receive, (Produce large amounts of small intracellular mediators/activate signaling proteins (adenylyl cyclase).

5 Transducer proteins: convert the signal into a different form (adenylyl cyclase).Gnaling.

6 Bifurcation proteins: spread signal from one pathway to another.

7 Integrator proteins: receive signals from 2++ pathways integrate them before relaying a signal onward.

8 Latent gene regulatory proteins: activated at the cell surface then migrate to nucleus to stimulate gene transcription.

Q. Write at least one function of scaffolds protein?

Many cells use scaffold proteins to enhance speed efficiency & specificity of response.

Q. What is abbreviation of GPK, GP, PP, GS?

glycogen phosphorylase kinase (GPK), glycogen phosphorylase (GP) phosphoprotein phosphatase (PP), glycogen synthase (GS)

Q. What is difficult between Agonists and antagonists?

Agonist (terbutaline) Mimic the function of a hormone by binding to its receptor and causing the normal response while Antagonist (β-blockers) Bind to the receptor but do not activate hormone induced effects.

Q. What are the causes of cAMP?

cAMP rise causes membrane proteins to permit a massive flow of water from the blood into the intestinal lumen, causing diarrhea (water loss) resulting in death from dehydration.

Q. How CML formed?

chronic myelogenous leukemia (CML) results from a chromosomal translocation between chromosome 9 and 22.

Q. What Is Bruton's tyrosine kinase,?

Bruton's tyrosine kinase, a protein whose mutation is associated with X-linked agammaglobulinemia.

Q. What is the function of Erythropoietin?

Erythropoietin stimulates erythrocyte precursor cells to survive, proliferate, and differentiate, is associated with Jak2

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Q. Write function of prolactin?

Prolactin stimulates breast cells to produce milk Activated STAT5 stimulates the transcription of genes that encode milk proteins.

Q. What do you know about TGF-β Receptor Family?

Consists of a large number of structurally related, secreted, dimeric proteins. During development, they regulate pattern formation and influence various cell behaviors, ie proliferation, differentiation, extracellular matrix production, and cell death. In adults, tissue repair and in immune regulation. Act through enzyme-linked receptors that are single-pass transmembrane proteins.

Q. How many receptors are required for signaling mentioned their name?

Two types of receptors are required for signaling Both type-I and type-II receptors, required for signaling.

Q. What is abbreviation of PDGF and EGF?

platelet-derived growth factor (PDGF) and epidermal

growth factor (EGF).

Q. Write characteristics of Jak stat signaling?

Jak-STAT Signaling.Increase antigen presentation of macrophages.Activate & increase lysosome activity macrophages.Cells to increase expression of MHC class I .Promotes adhesion and binding required for leukocyte migration.Promotes NK cell activity.Activate inducible Nitric Oxide Synthase.Jak-STAT one of the most direct signaling.Ligand binding induces receptor to dimerize.Binding brings Jaks close to cross-phosphorylate each other, increasing kinase activity. The Jaks then phosphorylate receptors, creating STAT docking sites

Q. Define antigens?

Substances that activate adaptive immune specific (AIS) are called antigens.

Q. Write names of five class of antibodies?

5 classes of antibodies, IgA, IgD, IgE, IgG, & IgM,

Q. Explain immunological memory?

Immunological Memory: The adaptive immune system, like the nervous system, can remember prior experiences. First exposure to an antigen A: Immune response appears after several days, rises rapidly and exponentially, and then, more gradually, declines. Secondary immune response: Lag period is shorter, and the response is greater indicating animal remembers its first exposure to antigen A. If

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animal is given a different antigen B instead of second injection of antigen A the response is typical primary not secondary, immune response.

Q. What are Cellular Basis of Immunological Memory?

When naïve lymphocytes are stimulated by their specific antigen, they proliferate and differentiate. Most become effector cells which function & then die, while others become long-lived memory cells.

Q. What is mechanism of class switching?

Class Switching: B-Cell initially produces IgM (antibody's heavy chain is encoded by the first constant region gene, the μ segment) molecules, which recognise specific antigenic determinant. When B cell becomes a plasma cell another deletion occurs in the cell's DNA, which position heavy-chain variable region gene next to a different constant region (γ, ε, or A)

Q. What are the types of MHC molecules?

There are two types of MHC molecules MHC class I and MHC class II with distinct functions.

Q. What is genetic diversity of MHC classes?

3 genes for class I proteins (HLA-A, B &C). 3 genes for class II proteins (HLA-DP, DQ &DR)6 genes each for both the classes.

Q. What happened after T cell activation?

Once a helper T cell is activated by an antigen-presenting cell to become an effector cell, it then help activate other cells.

Q. what is TH1 and TH2 explain them?

When activated by an antigen-presenting cell, a naïve helper T cell can differentiate into either of two distinct types of effectorhelper cell, called TH1 and TH2. TH1 cells mainly help activate macrophages and cytotoxic T cells, whereas TH2 cells mainly help activate B cells.

Q. Explain complement system?

Complement System: Most of the time, the binding of antibodies to antigen performs no useful function until and unless it can activate an effector mechanism.Complement system serves several effector roles.Complement system provides the actual protection from the response while the interaction of antibodies and antigen provides the specificity of the response.Put another way, antibodies "finger" the target, complement destroys it The complement system consists of some 30 proteins circulating in blood plasma.These proteins are synthesized mainly in the liver, and they account for about 5% of the globulin fraction of blood serum.Most of these are inactive until .they are cleaved by a protease which,

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in turn, converts them into a protease.Thusmany components of the system serve as the substrate of a prior component and then as an enzyme to activate a subsequent component.

Q. What do you know about proto oncogene? And write its classes?

A proto-oncogene is a normal gene that can become an oncogene, either after mutation or increased expression. Proto-oncogenes code for proteins that help to regulate cell growth& differentiation.

Classes : Growth Promoting Oncogenes, Growth-Inhibiting Cancer Suppressor Genes, DNA Repairing Genes, Apoptosis Genes.

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